

Amelogenesis

Clinical Considerations

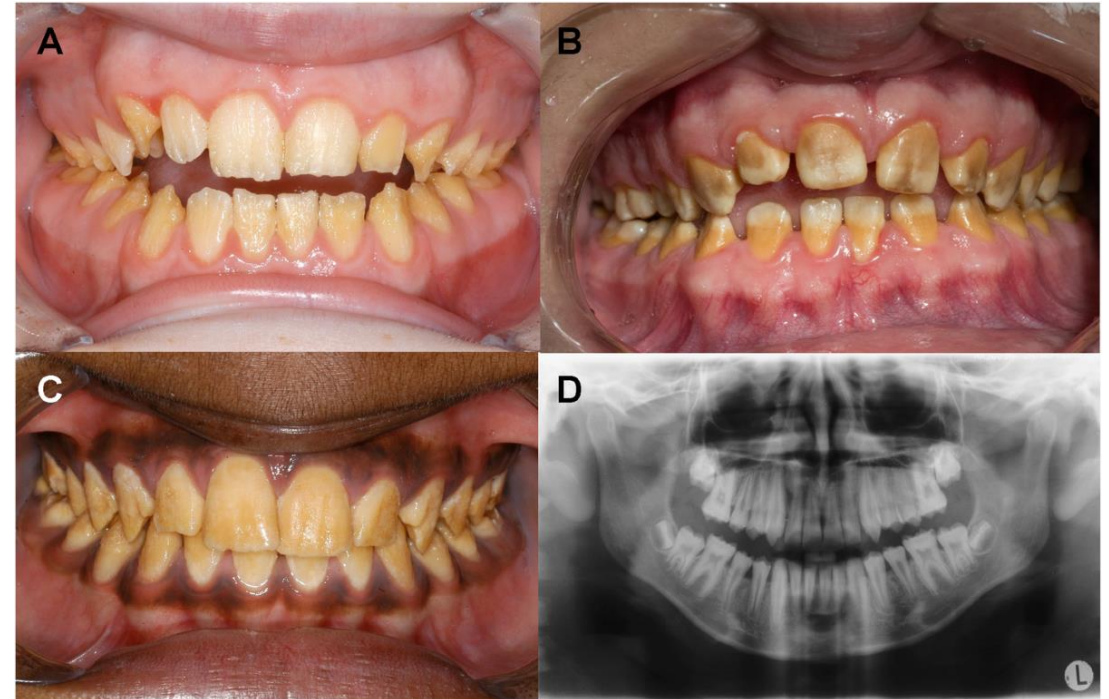
Clinical Use of Enamel Proteins

- **Title:** Emdogain® – Periodontal Regeneration
- **EMD** = Enamel Matrix Derivative
- Contains amelogenin
- Used in guided tissue regeneration
- Stimulates cementum and PDL regrowth



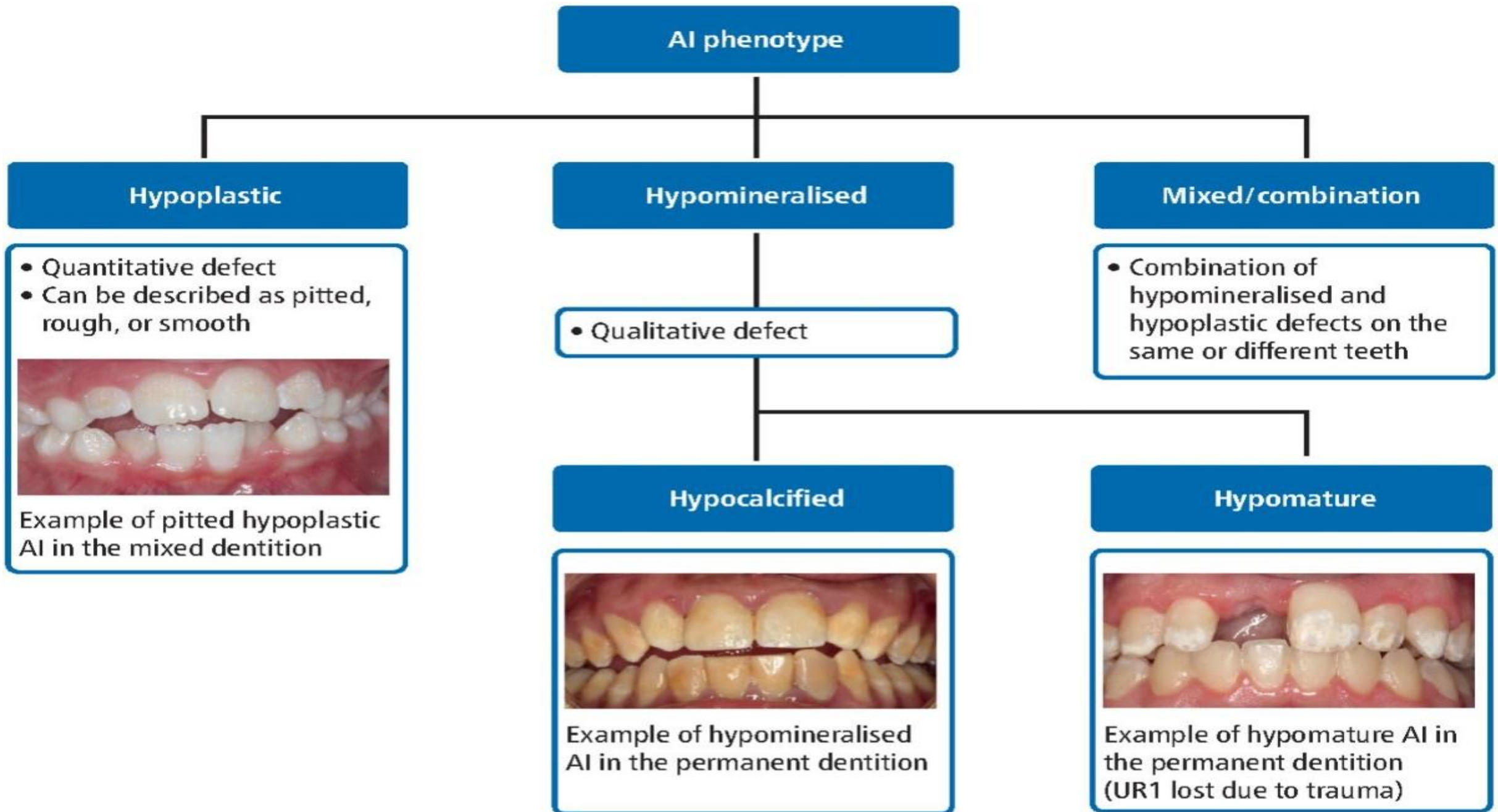
CLINICAL DISORDERS

- **Amelogenesis Imperfecta**
(Genetic Defects Affecting Only Enamel)
- **Prevalence:** 1 in 14,000 (approx)
- **Genes affected:** AMELX, ENAM, MMP20, KLK4
- **Dentin is NORMAL** (unlike other conditions)



Three Types of Amelogenesis Imperfecta

Type	Defective stage	Clinical appearance
Hypoplastic	Secretory (quantity)	Enamel is thin, hard, pitted
Hypocalcified	Early maturation	Enamel is soft, dark, chips easily
Hypomaturation	Late maturation	Enamel is normal thickness but mottled, brown



Dental Fluorosis

- **Cause:** >5 ppm fluoride during amelogenesis
- **Stage affected:** Maturation (ER stress in ameloblasts)
- **Appearance:** Diffuse white opacities → brown mottling → pitting



Tetracycline Staining

- **Mechanism:** Tetracycline chelates calcium and incorporates into crystal lattice
- **Appearance:** Yellow → brown → gray (under UV, it fluoresces)
- **Affects:** Enamel AND dentine
- **Prevention:** No tetracycline in children under 8 years



Molar Incisor Hypomineralization (MIH)

- **Definition:** Demarcated opacities in 1st permanent molars and incisors
- **Prevalence:** 2.4% to 40% (varies by population)
- **Aetiology:** Unknown – possible childhood fevers, dioxins, illnesses
- **Clinical problem:** Teeth crumble after eruption



Localized Trauma – Turner's Tooth

- **Title:** Trauma to Primary Tooth Affects Permanent Successor
- **Cause:** Periapical infection or trauma to primary tooth
- **Mechanism:** Inflammation damages developing permanent tooth germ
- **Result:** Localized hypoplasia (pit or white spot)



REVIEW & CONCLUSION

Feature	Amelogenesis	Dentinogenesis
Cell	Ameloblast (epithelial)	Odontoblast (mesenchymal)
Matrix	Amelogenin, no collagen	Collagen type I
Mineralization	Immediate (no predentin)	Delayed (predentin layer)
Process	Tomes process creates prisms	No prisms
Cell movement	Retreats from DEJ	Leaves process behind

- Defects: **AI** (genetic)
- **Fluorosis** (dietary)
- **MIH** (unknown)

Clinical Relevance

- **Caries susceptibility** – defects in enamel increase caries risk
- **Aesthetic restorations** – understanding enamel structure helps bonding
- **Local anesthesia** – enamel is acellular (no pain), but dentine is not
- **Orthodontics** – enamel must be etched (acid removes aprismatic layer)
- **Forensics** – amelogenin gene determines sex from teeth